Linear Algebra (MT-1004)

Homework#14

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Question 1

1. Compute the norm of the vector $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
2. Determine whether the vector $\mathbf{w} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}$ is a unit vector.
3. Normalize the vector $\mathbf{x} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$.
4. **Python Task:** Implement a Python function that returns the norm of a given vector.

```python
import numpy as np
def vector_norm(v):
    # Your code here
```